

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Operating Systems

COURSE CODE: CSA0401

COURSE OUTCOME COVERED

CO5: *Analyze the components and functionalities of Linux operating systems and virtualization environments to enhance their operational efficiency*

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

MOCK INTERVIEW

Code of Conduct:

I _G. Yashmita (192511032)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Yashmita

Annexure A

MOCK INTERVIEW

S.No.	Questions	Mark
1	How would you analyze the Linux file system structure and implement a long-term solution to prevent this issue?	3

2	You are given a 10 GB log file and need to extract all failed login attempts within 1 minute. Analyze the scenario and propose suitable Linux commands or shell techniques, along with performance optimization strategies, to accomplish the task efficiently.	5
3	A system is experiencing performance degradation due to a process consuming 95% of CPU resources. Analyze the situation and explain how you would identify, evaluate, and manage the process using appropriate Linux tools, without disrupting critical services.	5
4	Would you recommend a Type 1 or Type 2 hypervisor? Justify your answer based on performance, cost, and scalability.	3
5	A server with 16 GB of RAM must support multiple virtual machines. Analyze the scenario and propose an efficient resource allocation strategy to ensure optimal performance and balanced utilization across all VMs.	4
6	A Linux server runs multiple applications, where one application is negatively impacting the performance of others. Analyze the situation and propose a resource isolation strategy to ensure fair and controlled usage of CPU and memory among all applications.	5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	10 Responds accurately, promptly, and elaborates with relevant examples	10 Responds accurately, but elaborates with limited depth	10 Responds promptly, but with limited depth	10 Partial responses; need prompting

1. Linux file system structure

I would use `df -h` to check disk usage and `du -sh` to find which directories are consuming the most space. I would mainly check `/var/log` for large log files. For a long-term solution, I would use **logrotate** to automatically rotate, compress, and delete old logs. I would also set alerts when disk usage crosses 80%.

2. Extract failed login attempts from a 10 GB log

I would use commands like `grep` and `awk` to process the file line by line without loading the entire 10 GB file into memory.

```
grep "Failed login" logfile.log
```

For a specific one-minute interval, I would use `awk` to filter based on timestamps. For better performance, I would filter the timestamp and login keyword early and avoid unnecessary commands.

3. Process consuming 95% CPU

I would first identify the process using `top`, `htop`, or:

```
ps aux --sort=-%cpu
```

I would check its PID and determine whether it is a critical process. If needed, I would reduce its priority using `renice` or safely stop it using `kill`. I would avoid `kill -9` unless absolutely necessary.

4. Type 1 or Type 2 hypervisor

I would recommend a **Type 1 hypervisor** for servers because it runs directly on hardware. It provides **better performance, security, and scalability**. Type 2 is cheaper and suitable for personal computers and testing.

5. 16 GB RAM for multiple VMs

I would reserve some RAM for the host and distribute the remaining memory based on each VM's workload. For example, with 4 VMs, I could allocate

4 GB, 3 GB, 3 GB, and 2 GB, while keeping 4 GB for the host. I would monitor usage and adjust resources when required.

6. Resource isolation

I would use **Linux cgroups or systemd resource controls** to limit CPU and memory usage for each application.

For example:

CPUQuota = 50%

MemoryMax = 2G

This prevents one application from consuming all resources and ensures fair performance for other applications.

